

Springer Texts in Statistics

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An Introduction to Statistical Learning

with Applications in R

 Springer

ANALYTICS

Week 13-14

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*Examples Inspired By
SOA, CAS, Coaching
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Key Formulas This Week

- $z = b_0 + b_1x_1 + \dots + b_px_p$
- Logit: $g(\mu) = \ln \frac{\mu}{1-\mu}$, $\hat{\mu}_i = \frac{1}{1+e^{-z}}$
- Probit: $g(\mu) = N^{-1}(\mu)$, $\hat{\mu}_i = N(z)$
- Complementary Log-Log: $g(\mu) = \ln(-\ln(1 - \mu))$
- Odds = $\frac{p}{1-p}$ and $p = \frac{\text{Odds}}{1+\text{Odds}}$
- Bernoulli $l(\boldsymbol{\beta}) = \sum_i [y_i \ln \mu_i + (1 - y_i) \ln(1 - \mu_i)]$
- Bernoulli $D(\hat{\boldsymbol{\pi}}) = 2 \sum_i \left[y_i \ln \frac{y_i}{\hat{\mu}_i} + (1 - y_i) \ln \frac{1-y_i}{1-\hat{\mu}_i} \right]$
- Bernoulli Pearson $e_i = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1-\hat{\mu}_i)}}$
- Cumulative Logit Proportional Odds: $\text{logit}(F_Y(j)) = \ln \frac{F_Y(j)}{1-F_Y(j)} = \ln \frac{\pi_1 + \pi_2 + \dots + \pi_j}{\pi_{j+1} + \dots + \pi_c} = \alpha_j + \mathbf{x}'_i \boldsymbol{\beta}$, $\hat{F}_Y(j) = \frac{1}{1+e^{-(\alpha_j + \beta_1 x_1 + \dots + \beta_k x_k)}}$

Categorical Variables

- The goal of a **categorical dependent variable** model is to predict what _____
- **Nominal** categorical variables have unordered outcomes.
- **Ordinal** categorical variables have a natural ordering.

Binary Response

- A dependent variable with a binary response is discrete and not approximately normally distributed.
- Issues with using a linear probability model:
- **Poor fitted values:** $y \in [0,1]$ but $x'_i\beta$ could be any real value $(-\infty, \infty)$
- **Heteroscedasticity:** The variance of a binary response depends on the mean, which varies over observations.
- $Var(y_i) =$
- **Meaningless residuals:** Response is 0 or 1, and linear regression residuals assumes a continuous error distribution

Bernoulli Distribution

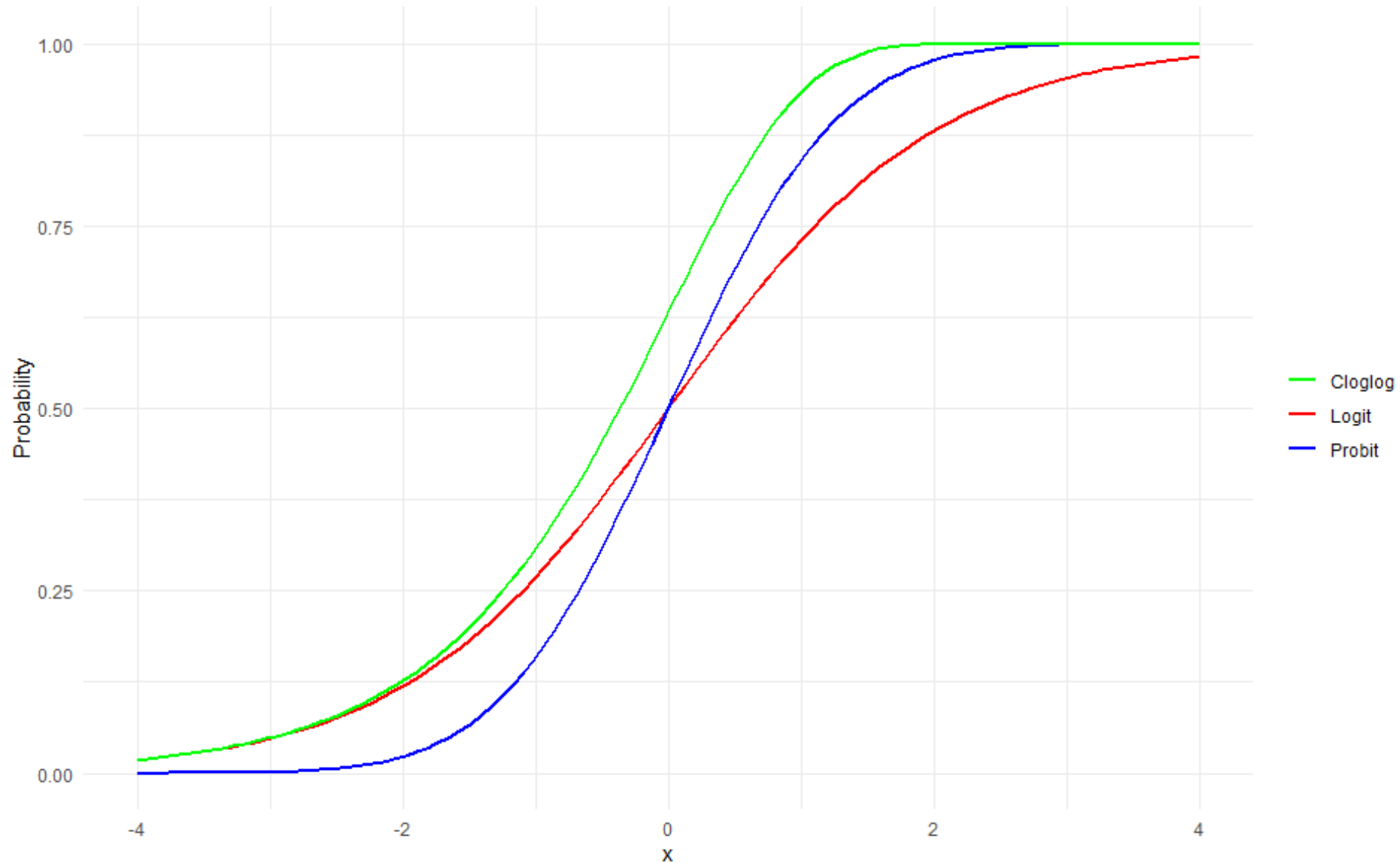
- $P(Y = y) = \pi^y (1 - \pi)^{1-y} =$
- in the linear exponential family format
$$f(y; \theta, \varphi) = e^{\left[\frac{y\theta - b(\theta)}{\varphi} + S(y, \varphi) \right]}$$
- $\theta =$
- $b(\theta) =$
- $\varphi = 1$ and $S(y, \varphi) = 0$
- $E[Y] = \pi$ and $Var(Y) = \pi(1 - \pi)$

Common Bernoulli Links

- Bernoulli distribution means are between $[0,1]$ so 3 common link functions for it are:
- Logit: $g(\mu_i) =$
- Probit: $g(\mu_i) = N^{-1}(\mu_i)$ using the inverse standard normal function (very similar to logit in practice)
- Complementary Log-log: $g(\mu_i) =$

Link Function Visual

Logit, Probit, and Cloglog Functions



Odds

- When y is binary $\{0,1\}$, then $\mu = p = P(y = 1)$.
- The odds of an event is the ratio of it happening vs. not
- $Odds = \frac{p}{1-p}$ and $p =$
- Log Odds: $\text{logit}(p) =$

Simple Odds Model

- If $\ln(\text{odds}) = \ln \frac{p}{1-p} = \beta_0 + \beta_1 x$ with x binary $\{0, 1\}$,
- $\text{odds} = \left\{ \right.$
- Odds ratio: $e^{\beta_1} =$

Prediction Summary

- $\mathbf{x}_i' \boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon$
- $\hat{\pi}_i = \hat{\mu}_i = \hat{p}_i$ are equivalent notation

Link Function	Solved Function
$\ln \frac{\pi_i}{1 - \pi_i}$	Logit: $\pi_i = \frac{e^{x_i' \beta}}{1 + e^{x_i' \beta}}$
$\Phi^{-1}(\pi_i)$	Probit: $\pi_i = \Phi(x_i' \beta)$
$\ln(-\ln(1 - \mu_i))$	CLL: $\pi_i = 1 - e^{-e^{x_i' \beta}}$

- $Var(\hat{\pi}_i) = \hat{\pi}_i(1 - \hat{\pi}_i)$

Concept Check #1

- I. The logistic and probit regression models circumvent the severe multicollinearity issues of linear models.
- II. The probit function is represented by $\hat{\pi} = N^{-1}(z)$ where $N()$ is the standard normal CDF.
- III. The logit and probit functions result in substantially different models.
- IV. A logit link function transforms the linear model into only predicting values between 0 and 1.
- V. For positive z , the complementary log-log function is more similar to the probit than the logit function.
- VI. For the same $g(\mu)$, the complementary log-log link function will always result in the largest prediction.

1. Odds Illustration

- The probability an application for auto insurance is denied is modeled by a logistic model with an intercept ($\beta_0 = -2$) and 2 explanatory variables.
- # of prior accidents has $\beta_1 = 0.9$.
- Marriage status (base case unmarried) is binary with $\beta_2 = -0.7$
- A. Find the odds and probability of denial if for an unmarried applicant with a history of 2 accidents.
- B. Find the odds ratio for a married vs. unmarried applicant.

1. Odds Illustration 2

- $\ln \frac{p}{1-p} = \mathbf{x}'\boldsymbol{\beta} = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \varepsilon$
- Odds = $\frac{p}{1-p}$ and $p = \frac{\text{Odds}}{1+\text{Odds}}$
- $\hat{\pi}_i = \frac{e^{x_i'\boldsymbol{\beta}}}{1+e^{x_i'\boldsymbol{\beta}}} = \frac{1}{1+e^{-x_i'\boldsymbol{\beta}}}$

Example 1

- Of students taking Exam P, 60% are underclassmen and 40% are upperclassmen.
- $\frac{1}{3}$ of underclassmen pass P.
- $\frac{1}{2}$ of upperclassmen fail P.
- Determine the probability of passing P.
- Find the odds that someone who passed is an underclassman.

Example 1 Sol

- Odds = $\frac{p}{1-p}$ and $p = \frac{\text{Odds}}{1+\text{Odds}}$

Example 2

- A GLM of the form: $g(\mu) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ is used to predict the chance of a dating match.
- If $g(\mu) = 0.75$, find the probability under a logit vs. probit vs. complementary log-log link function assumption.

Example 2 Sol

- Logit: $g(\mu) = \ln \frac{\mu}{1-\mu}$, $\hat{\mu}_i = \frac{1}{1+e^{-z}}$
- Probit: $g(\mu) = N^{-1}(\mu)$, $\hat{\mu}_i = N(z)$
- Complementary Log-Log: $g(\mu) = \ln(-\ln(1 - \mu))$

Example 3

- A hypnotist constructs 2 separate logistic models.
- Model A uses a response of subject is hypnotized. The explanatory variable measuring subject sleep deprivation has a $b_1 = 0.05$, with a value of x_1 producing a probability of hypnosis of 2%.
- Model B uses a response of subject is not hypnotized. The same explanatory variable has a $b_1 = 2$ in this model, with a value of x_2 producing a probability of no hypnosis of 95%.
- Both models have the same intercept b_0 .
- If $\frac{x_1}{x_2} = 10$, solve for x_2 .

Example 3 Sol

- $\ln \frac{\pi_i}{1-\pi_i} = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$

Diagnostics

- Concepts such as Deviance, Max-scaled R^2 , Pseudo R^2 , Likelihood Ratio test work the same way as they do for GLMs.
- $d_{i,Deviance} = \text{sign}(y_i - \hat{\mu}_i)\sqrt{D}$
- Standardized $d_{si} = \frac{d_i}{\sqrt{1-h_{ii}}}$
- $R^2_{maxscaled} = \frac{1 - e^{-\frac{2(l_0 - l)}{n}}}{1 - e^{-\frac{2l_0}{n}}}$, $R^2_{pseudo} = \frac{l_{\hat{\theta}} - l_0}{l_{SAT} - l_0}$
- $LRT = 2(l_a - l_0) = D_0 - D_a \sim \chi_q^2$

Bernoulli Deviance

- Bernoulli $l(\boldsymbol{\beta}) =$
- Bernoulli $D(\hat{\boldsymbol{\pi}}) =$

Bernoulli Pearson Residual

- Pearson residuals $e_{i,Pearson} = \frac{y_i - \hat{\pi}_i}{\sqrt{\text{Var}(y_i)}}$
generally, but for logistic GLMs simplifies to:

- $e_i =$

Ordinal Response

- Ordinal dependent variables have some natural ordering, so models are based on cumulative probabilities.
- $P(y \leq j) = \pi_1 + \pi_2 + \cdots + \pi_j = F_Y(j)$

Proportional Odds Model

- In a proportional odds cumulative logit model,
- $\text{logit}(F_Y(j)) = \alpha_j + \mathbf{x}_i' \boldsymbol{\beta} = \alpha_j + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}$
for $j = 1, 2, \dots, c - 1$
- The intercept term varies by _____
- α_j are _____
- Cumulative probit model works identically, using the probit link function.

Proportional Odds Model 2

- β_j is the proportional change in the _____
- To calculate predicted probabilities,
- $$\hat{F}_Y(j) = \frac{1}{1 + e^{-(\alpha_j + \beta_1 x_1 + \dots + \beta_k x_k)}}$$
- Cumulative Odds =

Simple Proportional Odds

- Simplified 3 class example $1 < 2 < 3$ with two explanatory variables:
- $\text{logit}(F_Y(1)) = \ln \frac{\pi_1}{\pi_2 + \pi_3} = \alpha_1 + \beta_1 x_1 + \beta_2 x_2$
- $\text{logit}(F_Y(2)) = \ln \frac{\pi_1 + \pi_2}{\pi_3} =$
- $\hat{\pi}_1 = \frac{1}{1 + e^{-(\alpha_1 + \beta_1 x_1 + \beta_2 x_2)}}$
- $\hat{\pi}_1 + \hat{\pi}_2 = \frac{1}{1 + e^{-(\alpha_2 + \beta_1 x_1 + \beta_2 x_2)}}$
- $\hat{\pi}_1 + \hat{\pi}_2 + \hat{\pi}_3 = 1$

Proportional Odds Strategy

- Simplified 3 class example $1 < 2 < 3$ with two explanatory variables:
- $\text{logit}(F_Y(1)) = \ln \frac{\pi_1}{\pi_2 + \pi_3} = \alpha_1 + \beta_1 x_1 + \beta_2 x_2$
- $\text{logit}(F_Y(2)) = \ln \frac{\pi_1 + \pi_2}{\pi_3} = \alpha_2 + \beta_1 x_1 + \beta_2 x_2,$
- Note that $\ln \frac{F_Y(1)}{1 - F_Y(1)} - \ln \frac{F_Y(2)}{1 - F_Y(2)} =$

Proportional Odds Steps

- If parameters are available, first find π_1 from the first cumulative logit model.
- Find π_2 from $\pi_1 + \pi_2$ of the second cumulative logit model.
- Repeat until the last probability, which is found so that all probabilities sum to 1.
- If parameters are not given but certain probabilities are, use the difference of the models and the $\alpha_1 - \alpha_2$ as the bridge.

2. Proportional Illustration

- A proportional odds model is used to determine the likelihood of lung cancer (1 low, 2 medium, 3 high).
- The probability of Low risk for a Male Smoker is 7.58%.
- The intercept for Low+Medium is 2 higher than that for Low.
- Calculate the probability of high risk for a Male Smoker.

Parameter	β
Intercept 1	?
Intercept 2	??
Gender	
• Male	-0.5
• Female	0
Smoking	
• Yes	-5
• No	0
Gender x Smoking	-1

2. Proportional Illustration 2

- $\ln \frac{F_Y(j)}{1-F_Y(j)} = \ln \frac{\pi_1 + \pi_2 + \dots + \pi_j}{\pi_{j+1} + \dots + \pi_c} = \alpha_j + \mathbf{x}'_i \boldsymbol{\beta} = \alpha_j + \beta_1 x_{i1} + \dots + \beta_k x_{ik} \text{ for } j = 1, 2, \dots, c-1$
- $\hat{F}_Y(j) = \frac{1}{1 + e^{-(\alpha_j + \beta_1 x_1 + \dots + \beta_k x_k)}}$

Generalized Logit Model

- A generalized logit model models log odds of category j relative to a base category k for a nominal categorical response.
- $\text{logit} \left(\frac{\pi_{i,j}}{\pi_{i,k}} \right) = \mathbf{x}'_i \boldsymbol{\beta}$
- $\pi_{i,j} = \frac{e^{(\mathbf{x}'_i \boldsymbol{\beta}_j)}}{1 + \sum_{m \neq k} e^{(\mathbf{x}'_i \boldsymbol{\beta}_m)}}$ for all $j \neq k$
- $\pi_{i,j} = \frac{1}{1 + \sum_{m \neq k} e^{(\mathbf{x}'_i \boldsymbol{\beta}_m)}}$ for the base category $j = k$

Example 4

- A proportional odds cumulative logit model is used to model the rank of an employee (1 = Probational < 2 Average < 3 Exceptional).
- Two continuous explanatory variables, hours worked X_1 , and sales amount X_2 are used.
- Employee Wonder Woman has a 20% probability of being ranked probational.
- Employee Starlord has double the probability of Probational vs. Average and his probability of Probational is 6 times that of Exceptional.
- Calculate the probabilities of each rank for Wonder Woman.

Example 4 Sol

- $\ln \frac{F_Y(j)}{1-F_Y(j)} = \ln \frac{\pi_1 + \pi_2 + \dots + \pi_j}{\pi_{j+1} + \dots + \pi_c} = \alpha_j + \mathbf{x}'_i \boldsymbol{\beta} = \alpha_j + \beta_1 x_{i1} + \dots + \beta_k x_{ik} \text{ for } j = 1, 2, \dots, c - 1$
- $\hat{F}_Y(j) = \frac{1}{1 + e^{-(\alpha_j + \beta_1 x_1 + \dots + \beta_k x_k)}}$

Summary Check

- Bernoulli linear exponential decomposition, Binary link functions, mean-variance relationship
- Formulas for log-likelihood, deviance, max-scaled R^2 , pseudo- R^2 , Pearson residual, deviance residual, overdispersion parameter, likelihood ratio test
- How to predict using the proportional odds cumulative logit model